

Given $a = x_0 < \cdots < x_n = b$ and $\xi_i \in [x_{i-1}, x_i]$, the Riemann integral of a bounded function f on $[a, b]$ is the tag-independent limit, when it exists,

$$\int_a^b f(x) \, dx := \lim_{\max_i (x_i - x_{i-1}) \rightarrow 0} \sum_{i=1}^n f(\xi_i)(x_i - x_{i-1}).$$